Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 5 — Straight line graphs
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第5章 · 直线
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C05-R01 · P1 · Chapter 5

$A = (-2, 5)$ and $B = (4, -1)$.

(a) Find an equation of the line AB. (3)

(b) Find an equation of the perpendicular bisector of AB. (3)

(Total for Question 1 is 6 marks)

2. P1-C05-R02 · P1 · Chapter 5

The line L has equation $3x + 2y = 12$.

(a) Find

(i) the coordinates where L meets the coordinate axes, (2)

(ii) the area of the triangle bounded by L and the coordinate axes. (2)

(b) Find an equation of the line through $(1, -2)$ perpendicular to L. (3)

(Total for Question 2 is 7 marks)

3.

Three points are $A = (1, 2)$, $B = (4, 8)$ and $C = (7, k)$.

(a) Find k if A, B, C are collinear. (2)

(b) For this value of k, find

(i) the ratio $AB : BC$, (2)

(ii) the midpoint of AC. (2)

(Total for Question 3 is 6 marks)

4.

The line L passes through $(2, 7)$ and is parallel to $4x - y = 9$.

(a) Find an equation of L. (2)

(b) Find where L intersects $x + 2y = 19$. (3)

(Total for Question 4 is 5 marks)

5.

$A = (-3, 1)$, $B = (5, 1)$ and $C = (1, 7)$ are vertices of a triangle.

(a) Find the area of triangle ABC. (2)

(b) Find an equation of the median from C to AB and an equation of the altitude from C to AB. (3)

(Total for Question 5 is 5 marks)

6.

P1-C05-R06 · P1 · Chapter 5

A straight-line calibration relates temperature T °C to a sensor reading R. Readings $R = 2$ and $R = 8$ correspond to $T = 15$ and $T = 51$, respectively.

(a) Find T in terms of R. (3)

(b) Use the model to find the reading for $T = 33$ °C, and state the temperature represented by the intercept. (3)

(Total for Question 6 is 6 marks)

$P = (a, 3)$ and $Q = (5, 11)$. The gradient of PQ is 2.

(a) Find a. (2)

(b) Find the exact length of PQ. (2)

(Total for Question 7 is 4 marks)

8. P1-C05-R08 · P1 · Chapter 5

Line L_1 is $y = mx + 4$ and line L_2 is $2y + 3x = 10$. They are perpendicular.

(a) Find m. (2)

(b) Find their point of intersection. (3)

(Total for Question 8 is 5 marks)

9. P1-C05-R09 · P1 · Chapter 5

The vertices $A = (0, 0)$, $B = (6, 2)$ and $C = (8, 8)$ are consecutive vertices of a parallelogram $ABCD$.

(a) Find the coordinates of D. (2)

(b) Find the point of intersection of its diagonals, and verify that both diagonals have this midpoint. (3)

(Total for Question 9 is 5 marks)

The line L passes through $A = (-1, 4)$ and $B = (3, -2)$.

(a) Find the equation of L in the form $ax + by = c$, where a, b, c are integers. (3)

(b) Point P lies on L and has equal coordinates. Find

(i) the coordinates of P, (2)

(ii) the exact distance AP. (2)

(Total for Question 10 is 7 marks)

A triangle has vertices $A = (2, 1)$, $B = (8, 3)$ and $C = (4, 7)$.

(a) Show that the triangle is right-angled at C or explain why it is not. (3)

(b) Find an equation of the line through C perpendicular to AB. (3)

(Total for Question 11 is 6 marks)

$M = (3, -2)$ is the midpoint of AB, where $A = (-1, 5)$.

(a) Find B. (2)

(b) Find an equation of the perpendicular bisector of AB. (3)

(Total for Question 12 is 5 marks)

13.

P1-C05-R13 · P1 · Chapter 5

A line through $(0, 6)$ meets the positive x-axis at A. The triangle formed with the axes has area 15 square units.

(a) Find the coordinates of A. (2)

(b) Find the equation of the line and its gradient. (2)

(Total for Question 13 is 4 marks)

The line L has equation $(k+1)x + 2y = 8$, where k is real.

(a) Find k if L is parallel to $y = 3x - 1$. (2)

(b) For this value of k, find

(i) the y-intercept of L, (1)

(ii) where L meets $y = 0$. (2)

(Total for Question 14 is 5 marks)

15. P1-C05-R15 · P1 · Chapter 5

Two roads are modelled by $y = \frac{1}{2}x + 1$ and $y = -2x + 11$. Coordinates are measured in kilometres.

(a) Find the junction and show that the roads meet at right angles. (4)

(b) A depot is at $(0, 1)$ on the first road. Find the exact distance from the depot to the junction. (2)

(Total for Question 15 is 6 marks)

16. P1-C05-R16 · P1 · Chapter 5

$A = (1, -3)$ and $B = (7, 5)$. Point P lies on AB with $AP : PB = 1 : 2$.

(a) Find the coordinates of P. (2)

(b) Find an equation of the line through P parallel to $y = 2x + 7$. (3)

(Total for Question 16 is 5 marks)

17.

A point $P = (t, 2t + 1)$ is equidistant from $A = (0, 0)$ and $B = (4, 0)$.

(a) Find t. (3)

(b) Find P and its exact distance from A. (3)

(Total for Question 17 is 6 marks)

The line L passes through $(2, 3)$ and has x-intercept 6.

(a) Find an equation of L. (3)

(b) Find the y-intercept and the area of the triangle enclosed by L and the axes. (3)

(Total for Question 18 is 6 marks)

19.

The vertices of a rectangle are $A = (1, 1)$, $B = (5, 3)$ and $C = (2, 9)$ in consecutive order.

(a) Verify that AB and BC are perpendicular. (3)

(b) Find the fourth vertex D and the exact area of the rectangle. (3)

(Total for Question 19 is 6 marks)

Lines L_1 and L_2 are $2x - y = 3$ and $x + 2y = 4$.

(a) Find their intersection P. (2)

(b) Find the equation of the line through P and the midpoint of $A = (-2, 4)$ and $B = (6, 2)$. (3)

(Total for Question 20 is 5 marks)

A straight line passes through $(a, 0)$ and $(0, 2a)$, where $a > 0$.

(a) Find its gradient and write its equation in terms of a. (2)

(b) The line also passes through $(3, 4)$. Find a and the area of the triangle it forms with the axes. (4)

(Total for Question 21 is 6 marks)

$A = (-4, 2)$ and $B = (2, 6)$. Point $C = (5, k)$ lies on the line through B perpendicular to AB.

(a) Find k. (3)

(b) Find the exact length BC. (2)

(Total for Question 22 is 5 marks)

Point $P = (4, 7)$ is reflected in the horizontal line $y = 2$ to form Q.

(a) Find Q and an equation of the line PQ. (3)

(b) Find the point where the line through Q parallel to $y = 3x + 1$ meets the x-axis. (3)

(Total for Question 23 is 6 marks)

A machine costs £4200 when new. Its value is modelled by a straight line and is £1800 after 6 years.

(a) Find the model $V = at + b$, where t is age in years. (3)

(b) Find the age at which the model predicts zero value, and explain why the model should not be used for larger ages. (3)

(Total for Question 24 is 6 marks)

Line L is $y = -\frac{2}{3}x + 4$. Point $P = (3, 1)$ does not lie on L.

(a) Find an equation of the perpendicular from P to L. (2)

(b) Find the foot F of this perpendicular and hence the exact shortest distance from P to L. (4)

(Total for Question 25 is 6 marks)

The line joining $A = (1, 5)$ and $B = (7, 1)$ intersects the line $y = x$ at P.

(a) Find P. (3)

(b) Find the ratio $AP : PB$. (3)

(Total for Question 26 is 6 marks)

The line L_k has equation $y = kx + 2 - k$, where k is real.

(a) Show that every line L_k passes through the same point, and find this point. (3)

(b) Find the member of this family perpendicular to $y = 4x - 7$. (3)

(Total for Question 27 is 6 marks)

A triangle has $A = (0, 4)$, $B = (8, 4)$ and $C = (3, -2)$.

(a) Find equations of the median from C and the perpendicular bisector of AB. (4)

(b) Find where the median meets the x-axis. (2)

(Total for Question 28 is 6 marks)

29.

The lines $L_1 : y = 2x + a$ and $L_2 : y = -x + b$ meet at $(3, 4)$.

(a) Find a and b. (2)

(b) For these values of a, b, find

(i) the x-intercept of each line, (2)

(ii) the area of the triangle formed by the two lines and the x-axis. (2)

(Total for Question 29 is 6 marks)

Point $A = (2, 1)$ is reflected in the line $y = x + 2$ to give point B. The segment AB is perpendicular to the mirror line and its midpoint lies on the mirror line.

(a) Find the equation of AB and its intersection M with the mirror line. (3)

(b) Find B and the length of AB. (4)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C05-R01 · 6 marks

(a) · 3 marks

$y = -x + 3$.

斜率(−1−5)/(4+2)=−1；代一点得截距3。

M1	正确斜率
M1	用点求截距
A1	直线方程

(b) · 3 marks

$y = x + 1$.

中点(1,2)，垂直斜率1；点斜式y−2=x−1。

M1	中点
M1	垂直斜率
A1	方程

Question 2

P1-C05-R02 · 7 marks

(a) Find

(i) · 2 marks

$(4, 0)$ and $(0, 6)$.

分别令y=0与x=0。

B1	横截距
B1	纵截距

(ii) · 2 marks

12 square units.

截距形成直角三角形，面积为4×6/2。

M1	正确底高面积式
A1	面积

(b) · 3 marks

$y = \frac{2}{3}x - \frac{8}{3}$.

L斜率−3/2，故垂线斜率2/3，过点求截距。

M1	负倒数斜率
M1	点斜式
A1	最终方程

Question 3

P1-C05-R03 · 6 marks

(a) · 2 marks

$k = 14$.

AB斜率2；BC斜率(k−8)/3必须等于2。

M1	用斜率相等
A1	k值

(b) For this value of k, find

(i) · 2 marks

$1 : 1$.

两段横增量都3且在同一直线上，或两长度均$3\sqrt{5}$。

M1	比较坐标增量或长度
A1	比值

(ii) · 2 marks

$(4, 8)$.

两坐标平均，恰好为B。

M1	正确平均坐标
A1	中点

Question 4

P1-C05-R04 · 5 marks

(a) · 2 marks

$y = 4x - 1$.

给定线斜率4，平行线同斜率，代(2,7)得截距−1。

M1	正确平行斜率
A1	方程

(b) · 3 marks

$(\frac{7}{3}, \frac{25}{3})$.

代y=4x−1得9x−2=19，x=7/3；回代得25/3。

M1	正确联立消元
A1	x
A1	y

Question 5

P1-C05-R05 · 5 marks

(a) · 2 marks

24 square units.

AB水平，底长8；C到AB竖直距离6，面积24。

M1	正确底与垂直高
A1	面积

(b) · 3 marks

Both are $x = 1$.

AB中点(1,1)，与C有相同横坐标；AB水平，其垂线也为过C的竖直线。

M1	求AB中点
A1	中线方程
A1	高线方程及方向

Question 6

P1-C05-R06 · 6 marks

(a) · 3 marks

$T = 6R + 3$.

斜率(51−15)/(8−2)=6，代入一对数值求截距3。

M1	用两读数求斜率
M1	求截距
A1	模型

(b) · 3 marks

$R = 5$; intercept represents 3 °C at $R = 0$.

33=6R+3得R=5；截距指读数零时的模型温度。

M1	反求读数
A1	R值
B1	截距解释

Question 7

P1-C05-R07 · 4 marks

(a) · 2 marks

$a = 1$.

8/(5−a)=2给5−a=4。

M1	正确斜率关系
A1	a

(b) · 2 marks

$4\sqrt{5}$.

坐标差4与8，勾股长度$\sqrt{80}$。

M1	正确平方距离
A1	最简根式

Question 8

P1-C05-R08 · 5 marks

(a) · 2 marks

$m = \frac{2}{3}$.

L2斜率−3/2，垂直斜率负倒数为2/3。

M1	正确L2斜率
A1	m

(b) · 3 marks

$(\frac{6}{13}, \frac{56}{13})$.

2x/3+4=5−3x/2，乘6得13x=6；回代。

M1	正确联立
M1	解精确横坐标
A1	交点

Question 9

P1-C05-R09 · 5 marks

(a) · 2 marks

$(2, 6)$.

平行四边形对边相同位移，D=A+(C−B)。

M1	使用相等位移
A1	D坐标

(b) · 3 marks

$(4, 4)$; midpoint of AC and of BD both $(4, 4)$.

AC平均为(4,4)；BD平均为((6+2)/2,(2+6)/2)。

M1	求一条对角线中点
M1	独立求另一中点
A1	一致结论

Question 10

P1-C05-R10 · 7 marks

(a) · 3 marks

$3x + 2y = 5$.

斜率$-6/4=-3/2$；用点斜式消去分母。

M1	正确斜率
M1	使用已知点
A1	整数系数式

(b) Point P lies on L and has equal coordinates. Find

(i) · 2 marks

$(1, 1)$.

令y=x代入3x+2x=5。

M1	联立y=x
A1	P

(ii) · 2 marks

$\sqrt{13}$.

A到P坐标差2、-3，长度$\sqrt{13}$。

M1	两点距离
A1	根式

Question 11

P1-C05-R11 · 6 marks

(a) · 3 marks

It is not right-angled at C: $m_{CA} = 3$ and $m_{CB} = -1$, with product $-3 \neq -1$.

分别计算两条过C边斜率，乘积不为−1，所以不垂直。

M1	正确两个斜率
M1	应用垂直条件
A1	明确否定结论

(b) · 3 marks

$y = -3x + 19$.

AB斜率2/6=1/3，故所求斜率−3，过(4,7)。

M1	AB斜率
M1	垂直斜率及过点
A1	方程

Question 12

P1-C05-R12 · 5 marks

(a) · 2 marks

$B = (7, -9)$.

由(A+B)/2=M，得B=2M−A。

M1	中点方程
A1	B

(b) · 3 marks

$y + 2 = \frac{4}{7}(x - 3)$.

AB斜率−14/8=−7/4；垂直平分线过M且斜率4/7。

M1	AB斜率
M1	垂线斜率并用中点
A1	方程

Question 13

P1-C05-R13 · 4 marks

(a) · 2 marks

$A = (5, 0)$.

面积$(1/2)\times$OA$\times 6$=15给OA=5；正x轴确定符号。

M1	面积关系
A1	A坐标

(b) · 2 marks

$y = -\frac{6}{5}x + 6$; gradient $-6/5$.

连接截距$(0,6)$和$(5,0)$，斜率$-6/5$。

M1	正确斜率
A1	完整方程

Question 14

P1-C05-R14 · 5 marks

(a) · 2 marks

$k = -7$.

L斜率−(k+1)/2=3，求k。

M1	正确参数斜率
A1	k

(b) For this value of k, find

(i) · 1 marks

$(0, 4)$.

令x=0，2y=8。

B1	截距

(ii) · 2 marks

$(-\frac{4}{3}, 0)$.

L为−6x+2y=8，令y=0。

M1	正确代入参数及y=0
A1	横截距

Question 15

P1-C05-R15 · 6 marks

(a) · 4 marks

Junction $(4,3)$; gradient product $(1/2)(-2)=-1$.

联立得2.5x=10；两个非竖直斜率乘积−1意味着垂直。

M1	联立
A1	交点
M1	梯度乘积
A1	垂直结论

(b) · 2 marks

$2\sqrt{5}$ km.

坐标差4和2，距离$\sqrt{20}$。

M1	正确两点距离
A1	精确距离与单位

Question 16

P1-C05-R16 · 5 marks

(a) · 2 marks

$(3, -\frac{1}{3})$.

从A走全位移的1/3，即(1,−3)+(2,8/3)。

M1	正确使用三分之一位移
A1	两坐标

(b) · 3 marks

$y = 2x - \frac{19}{3}$.

斜率2，代P求截距−1/3−6。

M1	采用相同斜率
M1	使用P
A1	方程

Question 17

P1-C05-R17 · 6 marks

(a) · 3 marks

$t = 2$.

PA²=t²+(2t+1)²；PB²=(t−4)²+(2t+1)²；相等后得t²=(t−4)²。

M1	建立等距离平方
M1	正确消去共同项
A1	t

(b) · 3 marks

$P = (2, 5)$; distance $\sqrt{29}$.

把t=2代回坐标，再用勾股求长度。

M1	回代求点
A1	点坐标
A1	距离

Question 18

P1-C05-R18 · 6 marks

(a) · 3 marks

$y = -\frac{3}{4}x + \frac{9}{2}$.

另一点为(6,0)，斜率−3/4，代入求截距9/2。

M1	正确识别截距点
M1	求斜率
A1	方程

(b) · 3 marks

$(0, \frac{9}{2})$; area $\frac{27}{2}$ square units.

截距9/2，三角形面积6×(9/2)/2。

B1	纵截距
M1	底高面积式
A1	面积

Question 19

P1-C05-R19 · 6 marks

(a) · 3 marks

$m_{AB} = 1/2$, $m_{BC} = -2$, product -1.

用两段的纵差除横差，乘积检验。

M1	求两个斜率
M1	应用乘积条件
A1	垂直结论

(b) · 3 marks

$D = (-2, 7)$; area 30 square units.

D=A+C−B；AB=$\sqrt{20}$，BC=$\sqrt{45}$，矩形面积=$\sqrt{900}$=30。

M1	正确第四点
M1	求两垂直边长度
A1	面积

Question 20

P1-C05-R20 · 5 marks

(a) · 2 marks

$P = (2, 1)$.

y=2x−3代入x+2y=4给5x=10。

M1	正确消元
A1	两坐标

(b) · 3 marks

$x = 2$.

AB中点为(2,3)，与P同横坐标，所以直线竖直，不能用有限斜率表达。

M1	求中点
M1	识别竖直线
A1	方程

Question 21

P1-C05-R21 · 6 marks

(a) · 2 marks

Gradient -2; $y = -2x + 2a$.

斜率$(2a-0)/(0-a)=-2$，纵截距$2a$。

M1	处理参数斜率
A1	方程

(b) · 4 marks

$a = 5$; area 25 square units.

$4=-6+2a$给$a=5$；底a、高$2a$，面积$a^2=25$。

M1	代点求a
A1	a值
M1	底高关系
A1	面积

Question 22

P1-C05-R22 · 5 marks

(a) · 3 marks

$k = \frac{3}{2}$.

AB斜率2/3；BC斜率−3/2，故(k−6)/(5−2)=−3/2。

M1	求AB斜率及垂线斜率
M1	正确参数方程
A1	k

(b) · 2 marks

$\frac{3\sqrt{13}}{2}$.

坐标差3和−9/2，长度√(9+81/4)=√117/2。

M1	正确平方距离
A1	最简根式

Question 23

P1-C05-R23 · 6 marks

(a) · 3 marks

$Q=(4,-3)$; $x=4$.

中间高度2=(7+yQ)/2给yQ=−3，横坐标不变；连接线竖直。

M1	使用反射中点关系
A1	Q
B1	直线

(b) · 3 marks

$(5,0)$.

过Q斜率3，方程y=3x−15；令y=0得x=5。

M1	正确平行直线方程
M1	令y=0
A1	交点

Question 24

P1-C05-R24 · 6 marks

(a) · 3 marks

$V = 4200 - 400t$.

初值b=4200，年变化(1800−4200)/6=−400。

M1	斜率
B1	初值
A1	模型

(b) · 3 marks

10.5 years; it would predict negative monetary values.

令V=0解t=10.5；更大t使右式负，不能代表该机器通常意义上的价值。

M1	解零值年龄
A1	年龄
B1	具体局限解释

Question 25

P1-C05-R25 · 6 marks

(a) · 2 marks

$y = \frac{3}{2}x - \frac{7}{2}$.

垂线斜率3/2，代入P求截距。

M1	正确垂直斜率
A1	方程

(b) · 4 marks

$F = (\frac{45}{13}, \frac{22}{13})$; distance $\frac{3\sqrt{13}}{13}$.

联立两线得x=45/13,y=22/13；PF坐标差6/13和9/13，长度√117/13。

M1	联立求垂足
A1	垂足
M1	两点距离
A1	精确最短距离

Question 26

P1-C05-R26 · 6 marks

(a) · 3 marks

$P = (\frac{17}{5}, \frac{17}{5})$.

AB为2x+3y=17；令y=x，得5x=17。

M1	求AB方程
M1	与y=x联立
A1	P

(b) · 3 marks

$2:3$.

P横坐标距A为12/5，距B为18/5；共线段长度之比等于相应横差之比。

M1	比较同线段的横差
M1	正确两个差
A1	最简比

Question 27

P1-C05-R27 · 6 marks

(a) · 3 marks

All pass through $(1, 2)$, since $y = k + 2 - k = 2$ when $x = 1$.

写成y=k(x−1)+2，x=1时参数项消失，点与k无关。

M1	重排或寻找参数系数为零
A1	固定点
B1	独立于k的说明

(b) · 3 marks

$k = -\frac{1}{4}$, so $y = -\frac{1}{4}x + \frac{9}{4}$.

垂直要求k×4=−1；代回截距2−k。

M1	正确垂直参数
A1	k
A1	方程

Question 28

P1-C05-R28 · 6 marks

(a) · 4 marks

Median: $y = 6x - 20$; perpendicular bisector: $x = 4$.

AB中点M=(4,4)；CM斜率6。AB水平，垂直平分线过M且竖直。

M1	中点
M1	中线斜率及方程
A1	中线式
B1	垂直平分线

(b) · 2 marks

$(\frac{10}{3}, 0)$.

令6x−20=0。

M1	使用中线方程并令y=0
A1	交点

Question 29

P1-C05-R29 · 6 marks

(a) · 2 marks

$a = -2, b = 7.$

把交点分别代入4=6+a与4=−3+b。

M1	代点到两条线
A1	两参数

(b) For these values of a, b, find

(i) · 2 marks

$L_1 : (1, 0)$; $L_2 : (7, 0)$.

分别令y=0并解一次方程。

M1	对两线设零值
A1	两截距

(ii) · 2 marks

12 square units.

底为两横截距之差6，高为交点纵坐标4，面积12。

M1	正确底高
A1	面积

Question 30

P1-C05-R30 · 7 marks

(a) · 3 marks

$AB: y=-x+3;\ M=(\frac{1}{2},\frac{5}{2}).$

垂线斜率−1，过A得y=−x+3；与y=x+2联立。

M1	正确垂线
M1	联立
A1	中点M

(b) · 4 marks

$B=(-1,4);\ AB=3\sqrt{2}.$

B=2M−A=(−1,4)；两点坐标差−3、3。

M1	用中点反求B
A1	B
M1	两点距离
A1	长度